

University of Rwanda
College of Science and Technology
School of ICT
Department of Computer and Software Engineering
Year 3

Module Title: Mobile Application Systems and Design

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Date: 06th April, 2026

Lab 6: Push Notifications with Firebase Cloud Messaging (FCM)

Summary

(i) Observations

- Device token: Each device gets a unique FCM token generated at first launch. Firebase uses this token to identify and deliver notifications to the correct device.
- Foreground vs Background: When the app is open, notifications are not shown automatically - they must be handled manually and displayed via a dialog or UI update, whereas when the app is in the background, Android displays notifications automatically in the notification bar.
- Permissions: For Android 13+, notification permission must be explicitly requested. Otherwise, no notifications will be received if no permission is granted by the user.
- Configuration files: The "google-services.json" and "firebase_options.json" files contain Firebase credentials, and must be correctly placed for the app to connect to Firebase. Wrong Google syntax (Groovy vs Kotlin DSL) causes build errors.
- Internet connection: The device must have an active internet connection to receive push notifications since FCM relies on Google's servers for message delivery.

(ii) Key steps followed

Step 1. Project setup - Firstly, I created a Flutter project using "Push-notification-app". Secondly, I set up a Firebase project on the Firebase console, and registered the Android app using the package name "com.example.push-notification-app", then downloaded "google-services.json" and placed it in the "android/app" folder so that the app connects to the Firebase.

step 2: Configuring Gradle files - I added the Google Services plugin in "android/build.gradle.kts" (project-level) and in "android/app/build.gradle.kts" (app-level). This allows the app to read the firebase configuration file.

step 3: Installing required tools and firebase configuration - I installed Node.js, Firebase CLI and FlutterFire CLI. Then logged into Firebase, and ran "flutterfire configure" in git bash which automatically generated "lib/firebase_options.dart". I ~~then~~ added the

step 4: Adding dependencies - I added "firebase-core" and "firebase-messaging" to "pubspec.yaml" then ran "flutter pub get" to install them.

step 5: Writing the application code - The following were done in the code;

- Initialized firebase in "main()" using "Firebase.initializeApp()".
- Requested notification permission using "requestPermission()".
- Retrieved the device token using "FirebaseMessaging.instance.getToken()" and displayed the token on the screen so as to be copied.
- Used "FirebaseMessaging.onMessage.listen()" to handle foreground notifications.
- Used "FirebaseMessaging.onMessageOpenedApp.listen()" to handle background notifications.
- Registered a background handler using "@pragma('vm:entry-point')" annotation.
- Showed a popup dialog using "showDialog()" when a notification arrives.
- Displayed the notification content in the app UI using "setState()".

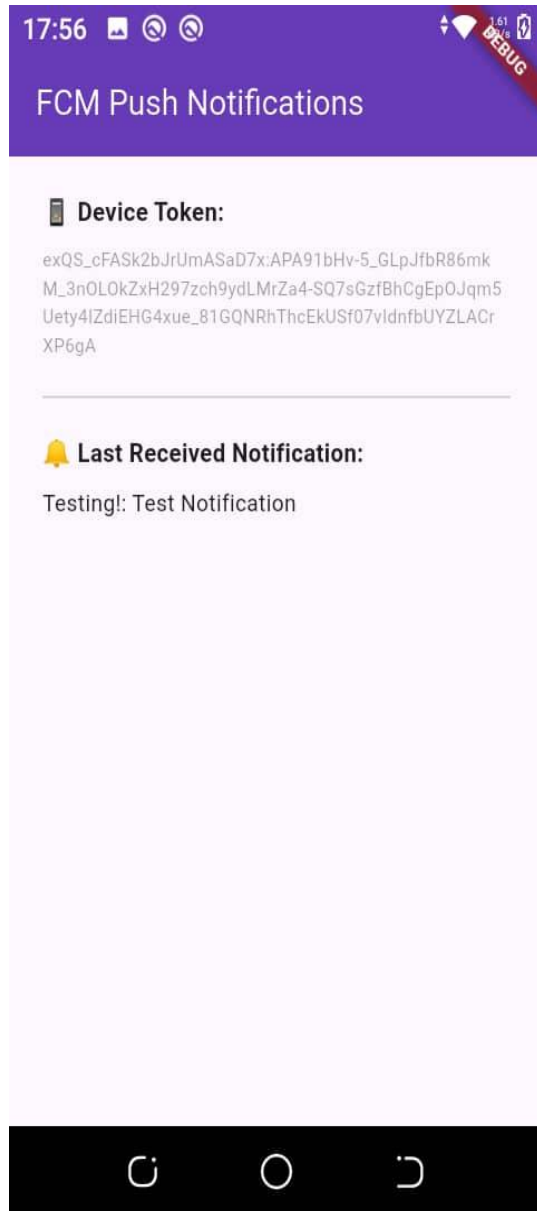
step 6: Tested on a real device - I enabled Developer options and USB debugging on my Android phone, connected the phone to my pc via USB and ran "flutter run" to deploy the app.

step 7: Sent Test Notification - I opened Firebase console → cloud Messaging and created a new campaign with title and body text then sent it using the device token. After this, I successfully received notifications both when the app was open and in the background.

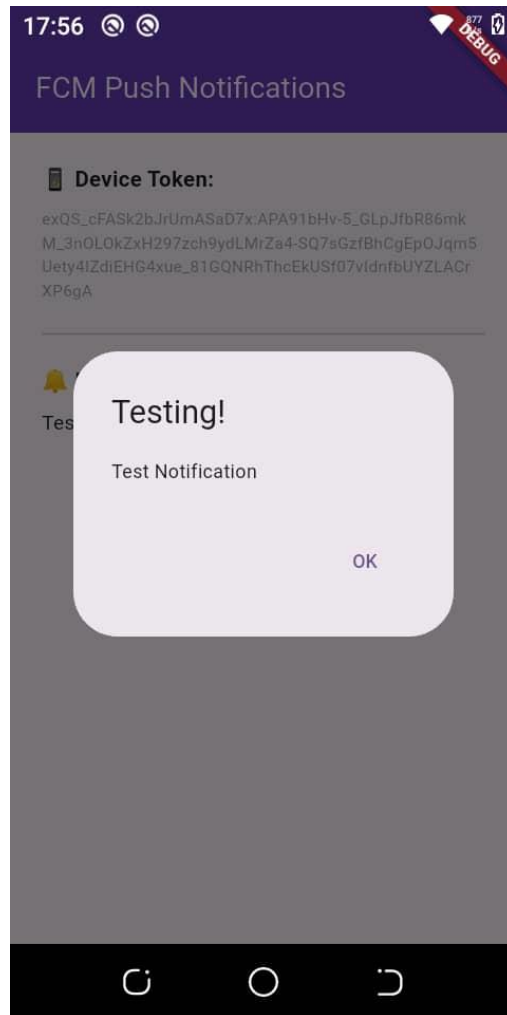
SCREENSHOTS.

i. Screenshots of the application.

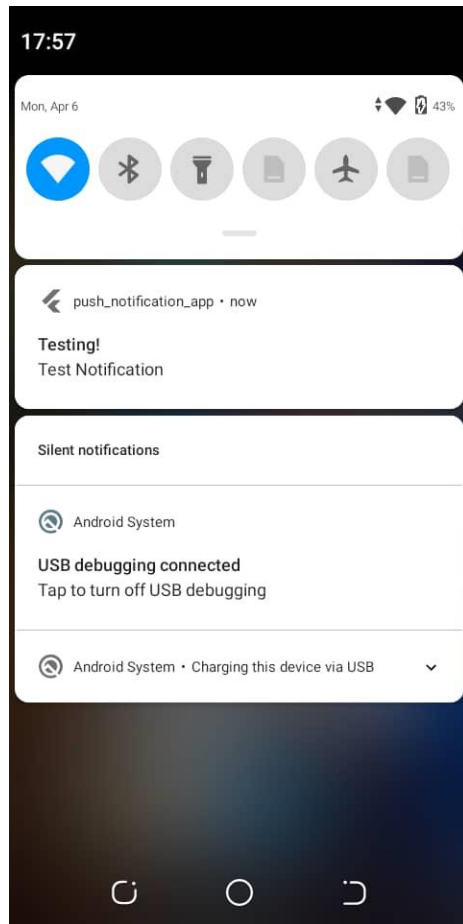
1. Inside the app UI



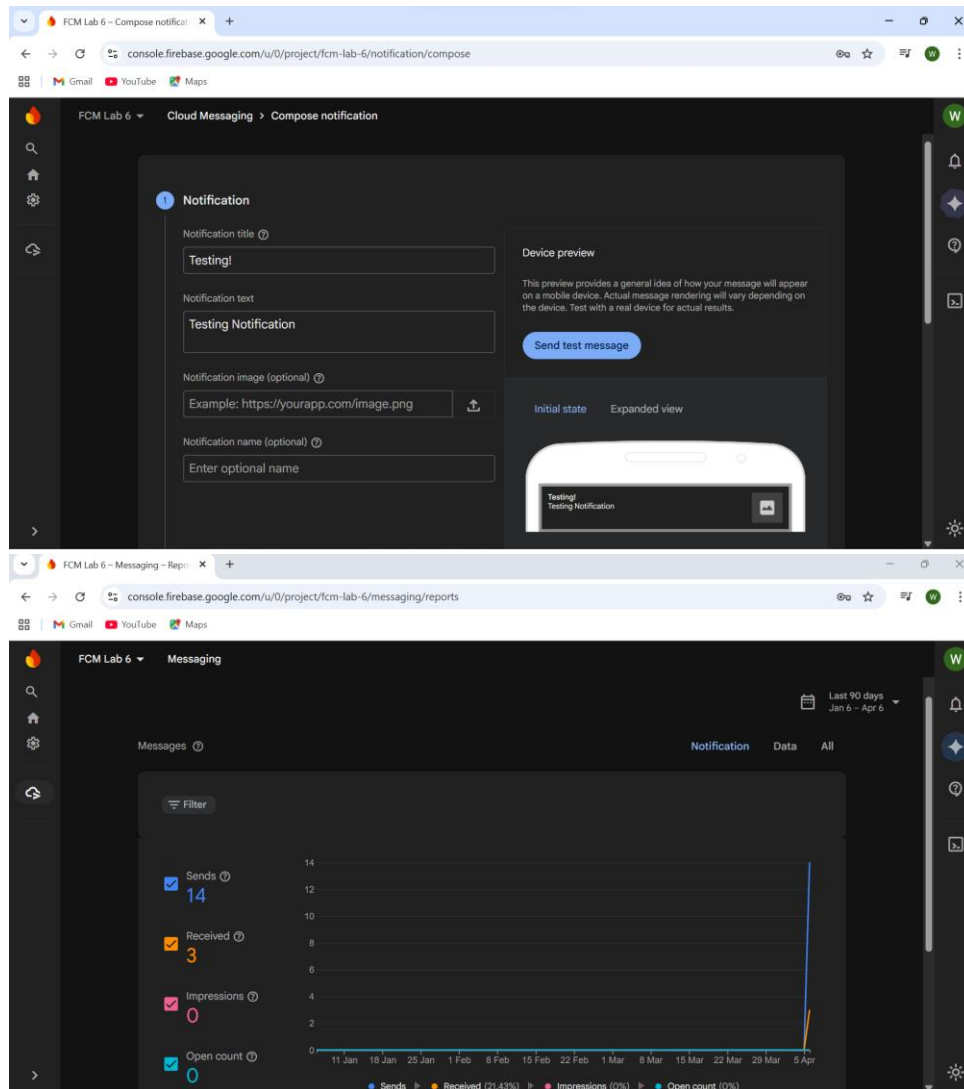
2. Popup



3. In the background (Notification bar)



ii. Screenshots of Firebase Cloud Messaging interface



- Sends are 14 because some could be send when I lost the connection to the device and could not be received.